

UNIT 11

OSCILLATIONS



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OSCILLATIONS:

"To and fro motion of a body about its position of equilibrium (Or mean position) is called oscillation."

Examples:

- Swinging of pendulum.
- Shivering of ring bell.
- Motion of a mass attached to a spring.

PERIODIC MOTION:

"The motion which repeats itself after a regular interval of time is called periodic motion."

Examples:

- Heartbeat of a person; period about 0.83 s for a normal person.
- Motion of earth around the sun; period one year.
- Swinging of bob of a simple pendulum.

SIMPLE HARMONIC FORCE:

"Vibratory motion in which the acceleration of the body is proportional to the displacement and always directed towards the equilibrium position is called simple harmonic motion."

$$a \propto -x$$

Examples of SHM:

- A simple pendulum.
- Projection of particle performing UCM.
- Spring mass system etc.

Conditions for simple harmonic motion:**• Restoring force:**

An elastic force is necessary in the system so that its motion is oscillatory.

• Inertia:

For simple harmonic motion system must have inertia.

• Acceleration and displacement:

The acceleration of the system at any point must be directly proportional to the displacement of the body from mean position.

• Direction of acceleration:

During simple harmonic motion direction of acceleration of the body is always remains towards equilibrium position.

• Frictionless system:

System should be friction less.

ACCELERATION OF OSCILLATING BODY (MOTION UNDER ELASTIC RESTORING FORCE):

Consider a body of mass 'm' attached at the end of an elastic spring. If the body is displaced from its equilibrium position and then released, it will start vibrating under the restoring force of the spring.

Let the body be at distance xxx from the equilibrium position at any instant, then the restoring force acting on it is given by:

$$F = -Kx \quad (1)$$

If a be acceleration of the body at displacement x then according to Newton's 2nd law of motion,

$$F = ma \quad (ii)$$

Comparing equation (i) and (ii),

$$Ma = -Kx$$

$$A = -\frac{K}{M}x \quad (iii)$$

It is an expression for the acceleration of a body executing S.H.M. In equation (iii) k/m is constant, therefore,

$$a = -(\text{constant})x$$

$$a \propto -xa$$

$$a \propto \text{displacement}$$

Since the acceleration produced by an elastic restoring force is proportional to the displacement and is directed towards the equilibrium position, therefore its motion is simple harmonic motion.

GRAPHICAL REPRESENTATION OF SHM:

To understand the simple harmonic motion graphically, we set up an experiment with an object attached to a spring. The object oscillates with a maximum displacement x_0 , from its equilibrium position.

At the same moment a pin projected up and fitted on a horizontal circular disc of radius $R = x_0$, is set into motion with constant angular speed ω . The two objects are set into motion at same instant.

Displacement – time (x-t) Graph:

At any time t the angular displacement of the pin is given by:

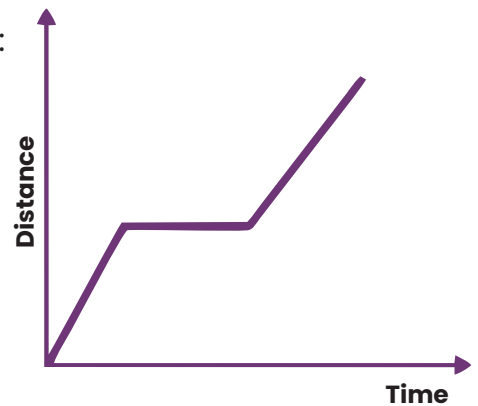
$$\theta(t) = \omega t$$

The displacement ' x ' of the shadow is just the projection of the radius ' r ' on to the x-axis. The particle starts on the x-axis at $x = x_0$ and moves through the angle θ in a time ' t '. Since the circular motion is uniform, the particle moves with constant angular speed ' ω '. Therefore, the angle has the value (in rad) of $\theta = \omega t$. The displacement ' x ' of the shadow is:

$$x = r \cos(\omega t)$$

$$\text{Where } \omega = \frac{2\pi}{T}$$

It is seen that value of x at different instants are recorded in the table below:

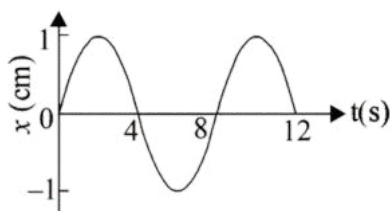


Time (t)	0	1/4T	1/2T	3/4T	T
Displacement $x = r \cos \frac{2\pi}{T}t$	r	-r	0	r	0

Velocity-time (v-t) graph:

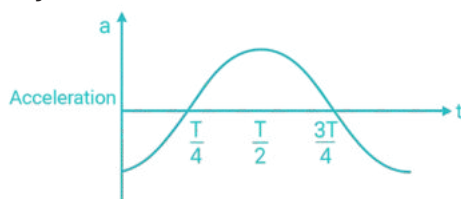
The shape of the curve for velocity-time relation is also the same as that of the displacement-time graph, but the displacement-time graph is one quarter ahead of the velocity-time graph. At time $t = 1/4T$, the mass is at equilibrium position O and this is the point where the velocity is maximum. The displacement ' x ' of the shadow is:

$$v = r\omega \sin(\omega t)$$



Acceleration-time (a-t) graph:

The relation between acceleration and time is described as, At $t = T$, the object is at O and there is no net force acting on it so its acceleration is zero, although the velocity is maximum at equilibrium position. As the object is displaced towards the right-hand side under the action of a net force, the restoring force will act towards the left, producing a negative acceleration in the object.



The acceleration-time graph is 180° out of phase with the displacement-time graph. This shows that:

$$a \propto -x$$

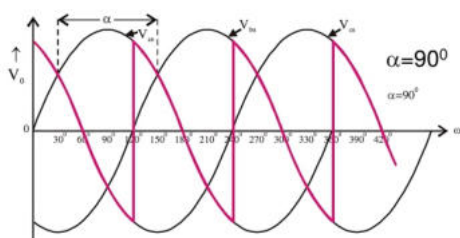
The acceleration 'a' of the shadow is:

$$A = r\omega^2 \cos(\omega t)$$

Comparison between displacement, acceleration, and velocity graphs:

The acceleration-time graph is 180° out of phase with the displacement-time graph, this shows that:

$$a \propto -x$$



PROJECTION OF PARTICLE PERFORMING UCM:

(Projection of particle moving in circular path and simple harmonic motion):

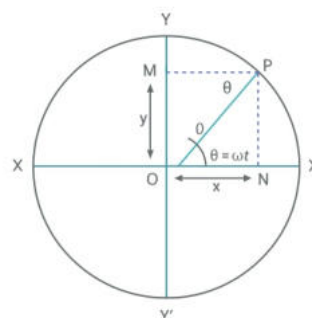
Let us consider a particle 'P' moving in a circle of radius 'r' with uniform angular velocity ' ω '. AB is the diameter of the circle and 'Q' is the projection of P along the diameter.

We start our observation at ' $t = 0$ '.

The centripetal acceleration of P is directed towards the center 'O' and is given by:

$$a_c = r\omega^2$$

(Thus, $v = r\omega$)



Since 'Q' is vibrating along the diameter, therefore its acceleration 'a' must be equal to the component of a_c parallel to the diameter i.e.,

$$a = a_c \cos \theta$$

$$a = r\omega^2 \cos \theta$$

(Thus, $a_c = r\omega^2$)

From figure $r \cos \theta = x$

Direction of x, From figure it is clear that a is directed towards mean position therefore,

$$a = \omega^2 (-x)$$

Since ' ω ' is constant,

$$a = -(\text{Constant}) x$$

$$a \propto -x$$

This proves that the motion of projection is simple harmonic motion.

Instantaneous velocity of projection:

As projection 'Q' is vibrating along the diameter, therefore its velocity 'V' must be equal to the component of V_p parallel to the diameter AB i.e.,

$$V = V_p \sin \theta$$

$$V = r\omega \sin \theta \quad (\text{Since } V_p = r\omega)$$

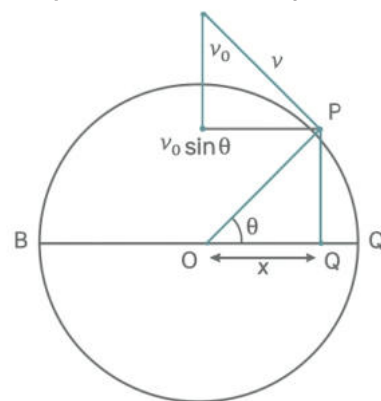
$$V = r\omega \sqrt{1 - \cos^2 \theta}$$

$$V = r\omega \left(1 - \left(\frac{x}{r}\right)^2 \right)$$

$$V = r\omega \left(\frac{r^2 - x^2}{r^2} \right)$$

$$V = \omega \sqrt{r^2 - x^2}$$

$$V = \omega (x_0^2 - x^2)$$



Maximum velocity:

Velocity of a particle performing SHM is maximum at its mean position i.e., at $x=0$:

$$V_{MAX} = \omega x_0$$

$$V_{MAX} = \omega x_0$$

Minimum velocity:

Velocity of a particle performing SHM is minimum at its extreme positions i.e., at $x = x_0$:

$$V_{MIN} = 0$$

RELATION BETWEEN LINEAR AND UNIFORM CIRCULAR MOTION:

An oscillating mass attached by spring and an object performing uniform circular motion has already been proven to execute simple harmonic motion.

For a body executing S.H.M, we know that,

$$a = -\frac{k}{m} x \quad (a)$$

And,

$$a = -\omega^2 x \quad (b)$$

By comparing equation (a) and (b),

$$-\omega^2 x = -\frac{k}{m} x$$

$$\omega = \sqrt{\frac{k}{m}}$$

TIME PERIOD OF SPRING MASS OSCILLATOR:

Since the object in the spring mass system and uniform circular motion both execute SHM, so by developing a relation between them,

$$T = \frac{2\pi}{\omega}$$

Where $\omega = \sqrt{\frac{k}{m}}$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

FREQUENCY OF SPRING MASS OSCILLATOR:

The number of oscillations in unit time is called frequency and it is the inverse of the time period,

$$F = \frac{1}{T}$$

$$F = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

VERTICAL SPRING MASS OSCILLATOR:

An oscillating mass on a vertical spring also exhibits SHM.

Suppose an object of mass 'm' and weight 'mg' is hung from the spring with spring constant 'K'. The spring is stretched downward under the gravity to a distance 'd' from the relaxed point A, and settled at 'O' the equilibrium position. Taking 'Y' axis in the upward direction, the net force acting on the mass is,

$$F_{\text{NET}} = \text{Spring force} - \text{Weight}$$

$$Kd = mg \quad (a)$$

If the object is raised from the equilibrium point 'O' to a position 'B' up to a distance of 'y', the spring force is less than Kd .

The spring force will be:

$$\text{Spring force} = K(d - y)$$

If the upward direction is taken as positive, then the net force acting on the mass at B is:

$$F_{\text{NET}} = K(d - y) - mg$$

$$F_{\text{NET}} = Kd - Ky - mg$$

From equation (a), $Kd = mg$ so:

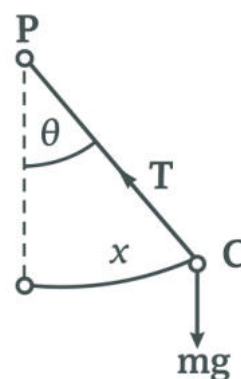
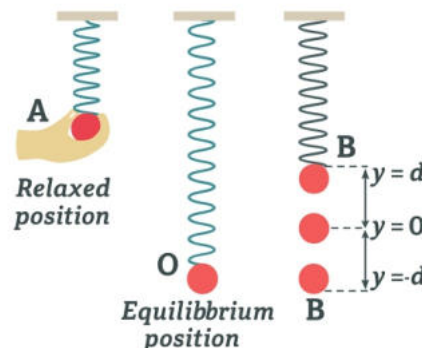
$$F_{\text{NET}} = mg - Ky$$

$$Ma = -ky$$

Where $\frac{K}{M}$ constant, so:

$$a = -(\text{constant})y$$

Hence, acceleration is directly proportional to vertical displacement and directed towards the equilibrium position. Therefore, the vertical spring mass system exhibits SHM.



SIMPLE PENDULUM:

A simple pendulum consists of a point mass suspended from a fixed point by a massless inextensible string of length 'l'. Consider a pendulum having a bob of mass 'm' suspended by a string of length 'l'. Let the bob be vibrating and at any instant its displacement be 'x' from the mean position.

Position. Suppose at any instant the pendulum makes an angle with the vertical axis. The following forces are acting on the bob of the pendulum:

Forces on the bob at 'C':

1. Weight: The weight of the bob 'mg' acting vertically downward.

2. Tension: The tension 'T' acting along the string.

Components of weight:

The 'mg' can be resolved into two rectangular components:

- $Mg \sin \theta$ acting perpendicular to the string.
- $Mg \cos \theta$ acting along the string.

Net force on the bob along the string:

Since the bob has no motion along the string, the component $mg \cos \theta$ must balance the tension 'T'. Therefore,

$$T = mg \cos \theta = 0$$

Net force on the bob perpendicular to the string:

The component $mg \sin \theta$ is the only force which brings the bob back to its equilibrium position. Thus, the restoring force acting on the bob at 'C' is:

$$F = -mg \sin \theta \quad (i)$$

If a' be the acceleration of the bob at 'C', then according to Newton's 2nd law of motion:

$$F = ma \quad (ii)$$

Comparing equation (i) and (ii) we get:

$$\begin{aligned} Ma &= -mg \sin \theta \\ a &= -g \sin \theta \end{aligned}$$

If θ is very small, then $\sin \theta \approx \theta$ so:

$$a = -g \theta$$

Using the relation,

$$\begin{aligned} S &= r \theta \\ \theta &= \frac{x}{r} \end{aligned}$$

Substituting the value of θ into the equation:

$$\begin{aligned} a &= -\frac{g}{l} x \\ a &= -(\text{Constant})x \Rightarrow \theta/l = \text{Constant} \end{aligned}$$

Above relation shows that acceleration of the bob is directly proportional to the displacement and directed towards the equilibrium position, hence the motion of the simple pendulum is SHM.

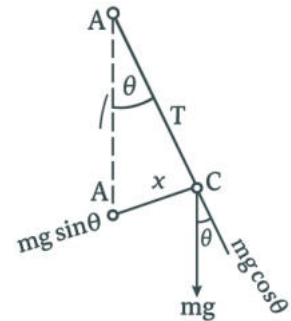
TIME PERIOD OF SIMPLE PENDULUM:

"Time required by the simple pendulum to complete one oscillation is called the time period."

To identify the angular frequency, we recall that equation:

For the case of pendulum,

$$a = -\omega^2 x$$



By comparing the above equations,

$$\omega^2 = \frac{g}{l}$$

$$\omega = \sqrt{\frac{g}{l}} \quad (i)$$

The time period of the body executing SHM is given by:

$$T = \frac{2\pi}{\omega}$$

Substituting the value of ω from equation (i):

$$T = \frac{2\pi}{\sqrt{\frac{g}{l}}}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Natural time period of simple pendulum:

"Time period of a given length pendulum at a specified location is constant and is known as the natural time period of the pendulum."

FREQUENCY OF SIMPLE PENDULUM:

"The number of oscillations completed by a simple pendulum in unit time is known as its frequency."

Frequency of the pendulum is the reciprocal of its time period:

$$f = \frac{1}{2\pi \sqrt{\frac{l}{g}}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$$

Use of Simple Pendulum:

We can find the value of 'g' with a simple pendulum, because both 'T' and 'L' can be directly measured. Proper precautions are taken; this method is the most accurate for the determination of 'g'.

PHYSICAL PENDULUM:

If the mass is uniformly distributed like arms swinging and legs moving of a man walking, then it is called a physical pendulum.

ENERGY OF A BODY EXECUTING S.H.M:

Consider a body of mass 'm' attached at one end of an elastic spring whose other end is fixed, and the motion of the body is S.H.M.

Potential Energy (U):

If the body is now pulled through a distance 'x' from its equilibrium position, the force acting on it at this point is given by:

$$F = kx$$

As force is directly proportional to 'x', therefore it is zero at the mean position because there $x = 0$ and at x distance.

The average force acting on the body during 'x' displacement is:

$$F = \frac{0 + kx}{2} = \frac{1}{2} kx$$

Work Done:

Work = Average force $\times$ displacement

$$W = \frac{1}{2} kx \cdot x = \frac{1}{2} kx^2$$

This work is stored in the body as potential energy:

$$U = \frac{1}{2} kx^2$$

Potential energy at mean position (Min P.E):

At mean position $x = 0$,

$$P.E = \frac{1}{2} k (0)^2$$

$$U_{\min} = 0$$

Potential energy at extreme position (Max P.E):

At extreme position $x = x_0$,

$$U_{\max} = \frac{1}{2} k (x_0)^2$$

KINETIC ENERGY:

The kinetic energy of a system at any instant is given by:

$$K.E = \frac{1}{2} mv^2 \quad (\text{iii})$$

But instantaneous velocity 'V' of the body at a distance 'x' from equilibrium position is,

$$V = \sqrt{\frac{K}{m}} \sqrt{x_0^2 - x^2}$$

Substituting the value of 'V' in equation (iii),

$$K.E = \frac{1}{2} m = \sqrt{\frac{K}{m}} \sqrt{x_0^2 - x^2}$$

$$K.E = \frac{1}{2} k \sqrt{x_0^2 - x^2}$$

K.E at mean position (Max K.E):

At mean position $x = 0$,

$$(K.E)_{\max} = \frac{1}{2} kx_0^2 - \frac{1}{2} k (0)^2$$

$$(K.E)_{\max} = \frac{1}{2} kx_0^2$$

K.E at extreme position (Min K.E):

At extreme position $x = x_0$,

$$(K.E)_{\max} = \frac{1}{2} kx_0^2 - \frac{1}{2} kx^2$$

$$(K.E)_{\max} = \frac{1}{2} kx_0^2 - \frac{1}{2} kx_0^2$$

$$(K.E)_{\max} = 0$$

TOTAL ENERGY:

Total energy at any point will be the sum of P.E and K.E.

At any displacement 'x':

Total energy = U + K.E. + K.E. + K.E.

$$E = \frac{1}{2} kx^2 + \frac{1}{2} kx_0^2 - \frac{1}{2} kx^2$$

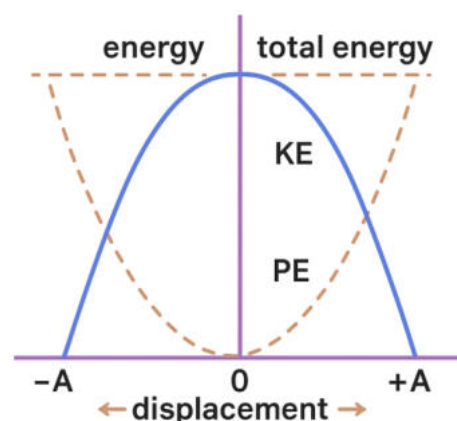
$$E = \frac{1}{2} kx_0^2 \quad (a)$$

At mean position:

$$E = U_{\min} + K.E_{\max}$$

$$E = 0 + \frac{1}{2} kx_0^2$$

$$E = \frac{1}{2} kx_0^2 \quad (b)$$

**At extreme position:**

$$E = U_{\max} + K.E_{\min}$$

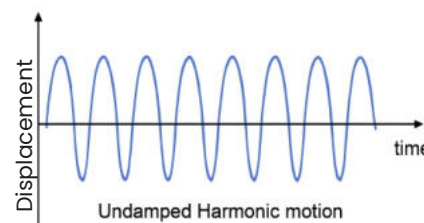
$$E = \frac{1}{2} kx_0^2 + 0$$

$$E = \frac{1}{2} kx_0^2 \quad (c)$$

By comparing equation (a), (b), and (c), it shows that **Energy of the System Executing Simple Harmonic Motion Remains Constant** throughout motion.

UNDAMPED OSCILLATION (Free Oscillation):

"Such type of oscillation in which amplitude of the particle executing SHM remains constant or increasing is called undamped oscillation."

**DAMPED OSCILLATION (Forced oscillation):**

"Such type of oscillation in which amplitude of particle executing SHM in each cycle is a little smaller than that of the previous one is called damped oscillation."

VEHICLE SHOCK ABSORBER:

A shock absorber used in a vehicle is the practical example of damped harmonic oscillation.

A shock absorber is a device that is integrated into the suspension system of a car or motorcycle. Its primary purpose is to dampen the oscillations caused by irregularities in the road surface when the vehicle encounters bumps. The shock absorber enables the spring to smoothly return to its equilibrium length without oscillating up and down.

RESONANCE:

"If the frequency of external driving force becomes equal to natural frequency or integral multiple of frequency, then the amplitude of the motion is maximum, this condition is called resonance."

MAGNETIC RESONANCE IMAGING (MRI):

Resonance has a very wide range of use in the medical science. MRI provides sufficient information about the patient's disease.

The strong magnetic field created by the MRI scanner causes the atoms in your body to align in the same direction. Radio waves are then sent from the MRI machine and move these atoms out of the original position. As the radio waves are turned off, the atoms return to their original position and send back radio signals. These signals are received by a computer and converted into an image of the part of the body being examined. This image appears on a viewing monitor.

Q – FACTOR:

"During damped oscillation, the ratio between stored energy and energy lost per oscillation is called Q-factor."

$$Q \text{ Factor} = \frac{\text{Energy Stored}}{\text{ENERGY LOST}}$$

Q factor is the dimensionless quantity.